



BrighterDays Mentor

OMDENA HYDERABAD CHAPTER, INDIA

Team 5



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Introduction

Social media is everywhere, for many, it is one of the first and last activities of the day. It is where time is passed, statuses posted and friendships are made. It is a place to stay in touch with friends, communicate with distant family and even find love. The world of technology is abuzz with advancements in large language models (LLMs), which are powering a new wave of applications. But amidst this progress, a crucial area remains relatively uncharted. The potential of LLMs to transform how we understand and address mental health on a global scale. One of the key benefits of using AI chatbots for mental healthcare is that they can provide personalised support and guidance. By using ML algorithms, these technologies can learn about an individual's unique needs and preferences, and tailor their responses accordingly.

Summary

The Omdena Hyderabad Local Chapter team embarked on a project to analyse social media data related to mental health using large language models (LLMs) and develop chatbot tools to support mental health management. Focusing on understanding global mental health trends, they collected diverse data sources, including Kaggle, Reddit, Facebook, and PDF documents, totaling over 1000 conversational data pairs and 60000+ entries. Preprocessing involves cleaning, encoding, and standardising the data. Model building is a crucial aspect of the project, involving the development and fine-tuning of mathematical or computational representations to analyse, predict, or understand various aspects of data. For sentiment analysis, DistilBERT was employed, achieving an accuracy of 90% precision for both depression and non-depression classes. KMeans Clustering was utilised for correlation analysis, revealing insights into the relationship between depression levels, anxiety, stress, gender, age, and education level. A Sentence Transformer model from Hugging Face facilitated pattern identification, enhancing data retrieval based on sentence similarity. A sophisticated RAG pipeline improved response quality, incorporating a parent-child retriever mechanism and the Mistral 7b instruct v0.2 model for response generation. Model fine-tuning via Structured Fine-Tuning (SFT) and Direct Preference Optimization (DPO) optimised system performance. Leveraging the Mistral-7B-Instruct-v0.2 model for both processes enhanced effectiveness in mental health analysis and support. VectorDB: Used ChromaDB to store vector representations of documents from CSV and PDF files, enabling quick retrieval for analysis. Integrated LlamaGuard for system safety, classifying user input as safe or unsafe. UI and integration efforts were meticulous, ensuring seamless performance. Sentence Transformers were used for embedding generation, managed by a loader script for text data transformation and insertion into the vector database. A conversation chain maintained continuity and utilised chat history for context. Prompt templates facilitated supportive conversations about mental well-being. A Chatbot Conversation Chain with BufferMemory enabled communication, integrating retrieval-augmented generation for relevant responses. Chatbot and recommendation system integration managed dialogue history effectively. Streamlit was employed for a user-friendly interface, allowing seamless interaction via web browsers. Model deployment on Hugging Face Spaces enabled easy hosting and sharing, with regular updates for optimization.

Task

[Omdena Hyderabad Local Chapter](#) team aims to use large language models (LLMs) to analyse massive amounts of data related to social media and mental health. This data could include social media posts, online surveys, and even news articles. By analysing this data, LLMs can identify patterns and trends that help understand the connection between social media use and mental well-being. The project also proposes developing chatbot tools powered by LLMs to help people manage their mental health, particularly in the context of social media.

Problem Statement

From the past two decades social media beheld a sporadic enhancement in quantity, quality and utility. As the body of an individual is nourished by the intake of necessary mineral elements obtained through nutrition, likewise the human mind is nurtured by the availability of nutrition for thoughts. Nowadays which is readily available through the advancement of technology. From the dawn of internet and social networking sites human resources of the contemporary world have become more social virtually but less practically. This virtual life is isolating the present man from other fellow beings thereby affecting his health (mental & physical) and overall balance. As a confronting population of the present era, the younger generation is experiencing an embryonic stage of life and is at higher risk of serious mental health problems. Younger generation of the present era is acting as active users of social media which has affinity towards the problems of mental health.

Goals and Objective

This project aims to utilise large language models (LLMs) and chatbot technology to gain a deeper understanding of global mental health trends. To achieve this, we will first develop a chatbot interface that allows users to anonymously share their mental health experiences and symptoms. The LLM will then be trained to analyse these chatbot conversations, identifying patterns in language use that correlate with specific mental health conditions. By analysing the LLM data, we can then identify trends in mental health concerns across different regions of the world.

Market Research

The objective of [market research](#) is to identify the needs and preferences of individuals regarding mental health services and support. Mental health is a state of mental well-being that enables people to cope with the stresses of life, realise their abilities, learn well and work well, and contribute to their community. Good mental health enables individuals to cope with life's challenges and contribute positively. Mental health problems rise by 13% annually, affecting 300 million with depression, 284 million with anxiety worldwide, and making suicide the fourth leading cause of death among 15-29-year-olds. Social media poses significant mental health challenges: 58% report negative impacts, 64% may experience anxiety disorders, 56% face depression due to electronic disconnection, 60% perceive others' lives as perfect, 59% worry about cyberbullying, 71% blame misinformation, and there's a 53%

higher likelihood of new ADHD symptoms among high school students with increased social media use.

Area of Focus

The area of focus for our research encompasses eight categories of mental health disorders, including Anxiety Disorders, Depression, Bipolar Disorder, Post-Traumatic Stress Disorder (PTSD), Schizophrenia, Eating Disorders, Disruptive Behaviour and Dissocial Disorders, and Neurodevelopmental Disorders. Out of the eight categories researched, our focus narrowed down to four specific areas: Depression/Loneliness, Attention Deficit Hyperactivity Disorder (ADHD), Self-esteem, and Anxiety Disorders.

Primary Focus: Among these, our primary focus lies on Depression. Depression is a widely recognized mental health condition affecting many individuals globally. By studying depression, we aim to uncover patterns and insights that can inform strategies for addressing mental health challenges in diverse global contexts, including the potential role of social media in exacerbating or mitigating depressive symptoms.

Data Collection

[Data collection](#) involved gathering pertinent information from diverse sources to understand the type of data beneficial in aiding individuals with mental health wellness. Sources included counselling websites and common inquiries in therapist forums. The goal was to acquire insights into coping strategies and support systems for managing depression.

Data Sources:

- [Kaggle](#): We obtained a portion of the data from Kaggle, a platform for datasets. This served as a foundation for our analysis.
- [Web Scraping](#) (Reddit and Facebook): Using Python, we scraped conversations from Reddit and Facebook, two popular social media platforms where discussions about mental health often occur.

Data Quantity: The collected dataset comprises 327 rows of data, each representing a conversation between a therapist and a user dealing with depression. This sample size provides a substantial basis for analysis.

Content Analysis: The conversations predominantly focus on topics related to coping with depression and strategies to alleviate its effects. Common themes explored include various coping mechanisms, therapeutic techniques, and lifestyle changes to manage depression effectively.

Textual Data: PDFs containing recommendations on dealing with depression were also obtained. These documents offer additional insights into evidence-based strategies for coping with and overcoming depression.

Privacy and confidentiality of individuals participating in the conversations were carefully maintained throughout the data collection process. Measures were taken to anonymize and de-identify personal information to uphold ethical standards.

We collected a comprehensive range of datasets, totaling over 1000 conversational data pairs, 60000+ entries in our balanced classification dataset, recommendation and intervention strategies extracted from reputable sources like the [American Psychiatric Association](#) and from [HelpingGuide.org](#), and more than 500 classification datasets sourced from university surveys, each enriched with specific features for analysis, encompassing topics such as depression and cyberbullying.

Data Preprocessing

In Machine Learning, data preprocessing plays a critical role in enhancing the quality of data to enable the extraction of meaningful insights. This process involves several essential steps:

Cleaning the Dataset: Removing odd special characters, eliminating duplicate entries, and handling missing values (NaN) to ensure data cleanliness and consistency.

Label Encoding: Replacing numerical classifications with corresponding labels to improve interpretability and facilitate analysis.

Data Integrity Assurance: Removing unknown labels to maintain data integrity and reliability, ensuring that only valid and recognized categories are included in the dataset.

Transformation and Standardization: Applying transformations as needed, including text modifications, to standardise formatting and rectify errors, thereby enhancing the consistency and reliability of the data.

Feature Selection: Selecting relevant columns based on their importance or relevance to the specific modelling task, refining the dataset to focus on key variables that contribute significantly to the analysis.

Utilising NLTK for Text Data Processing: In addition to traditional data preprocessing techniques, we leveraged the NLTK library to address text data processing requirements. NLTK offers a diverse range of tools and methods tailored for processing raw text data:

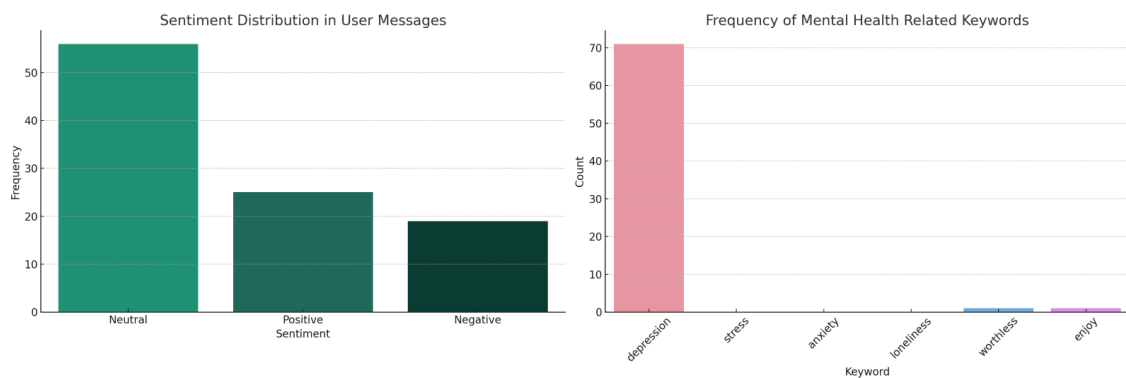
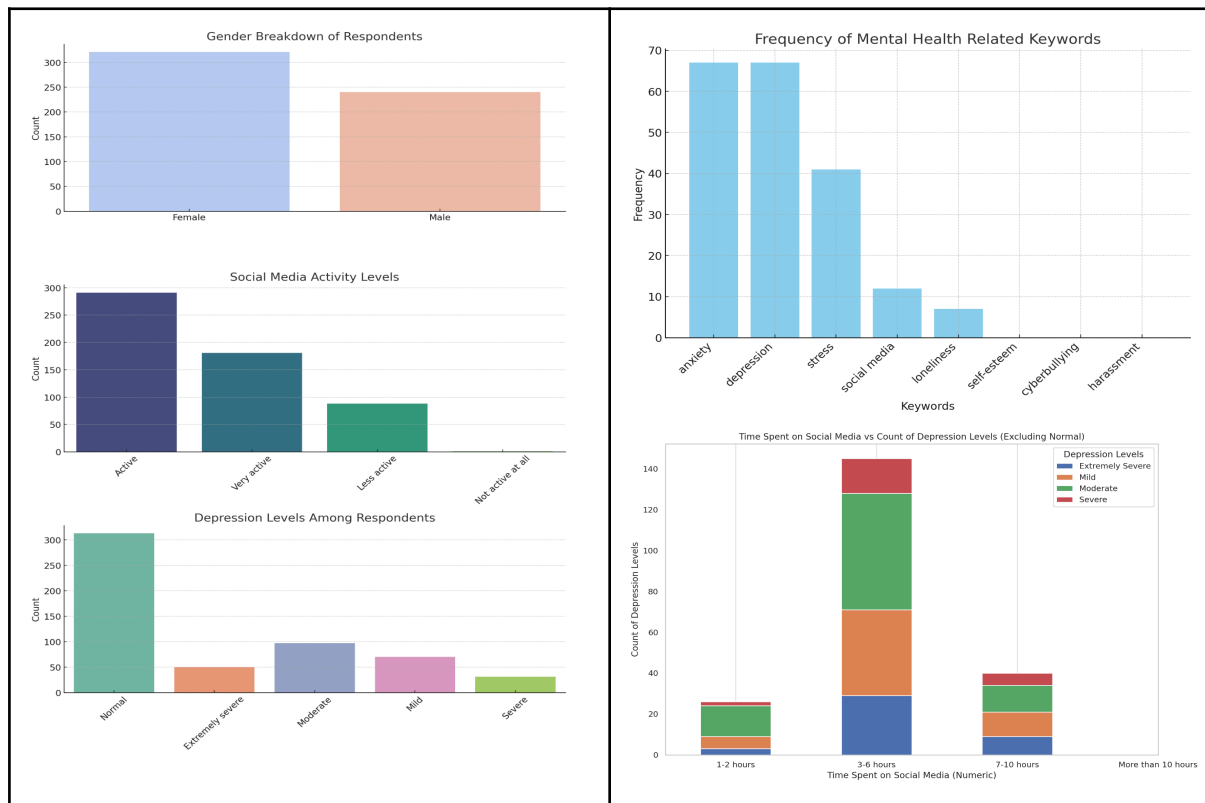
Tokenization: NLTK facilitates tokenization, enabling the segmentation of text into individual words or sentences, laying the groundwork for deeper analysis and understanding of textual data.

Data Visualization and Text Cleaning for Mental Health Analysis:

As part of our analysis, we employed various visualisation techniques and data cleaning methods to enhance the quality of our dataset and facilitate a deeper understanding of mental health levels across different sources:

Word Cloud and Bar Graph Visualisation: Utilising word clouds and bar graphs, we visually represented the levels of mental health derived from diverse data sources. These

visualisations provided insightful snapshots of mental health trends and variations, enabling clear interpretation and communication of findings.

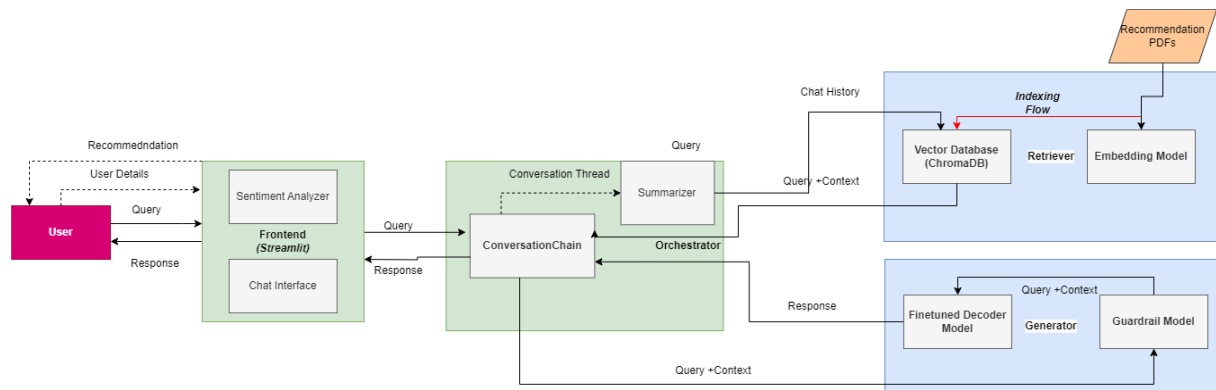


Text Cleaning with Regex: Given the presence of numerous acronyms and offensive language within the dataset, we recognized the importance of cleaning the text data to mitigate potential biases and ensure accurate analysis. Leveraging the regex library, we

implemented systematic text cleaning procedures to filter out acronyms and offensive language, thereby improving the quality and reliability of the dataset for subsequent analysis.

By integrating these visualisation techniques and text cleaning methodologies into our analysis pipeline, we were able to effectively navigate the complexities of the dataset and derive meaningful insights regarding mental health levels across different data sources. This comprehensive approach not only enhanced the robustness of our analysis but also facilitated clearer communication of findings to stakeholders.

Model Building



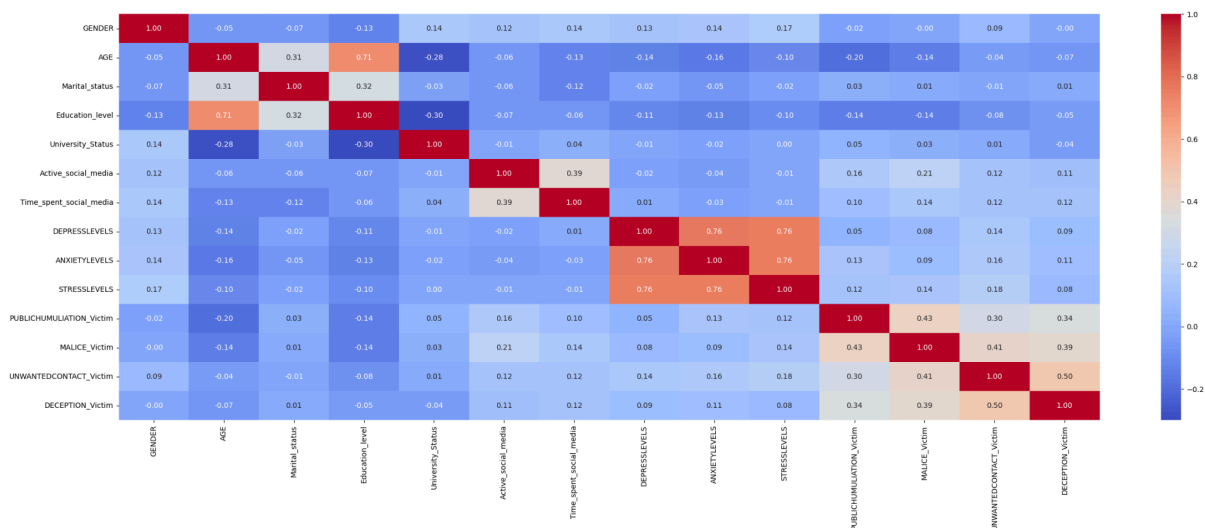
Model building is the process of developing mathematical or computational representations of real-world phenomena to analyse, predict, or understand various aspects of data.

In this context, we fine-tuned our model, which involves optimising a pre-existing model to better perform specific tasks. Specifically, we enhanced our model for Sentiment Analysis, Pattern Identification, and Correlation Analysis.

Sentiment Analysis: For sentiment analysis, we employed a text classification dataset featuring two columns: "text" and "label" (indicating whether the text reflects depression or not). Utilising [DistilBERT](#), we trained the model for approximately 20-30 epochs, achieving a remarkable accuracy of 90% in precision for both classes. In this setup, the input comprises summarised text, and the output predicts whether the text is associated with depression or not. The model loading process is facilitated using an auto tokenizer.

Correlation Analysis: For Correlation Analysis, we employed KMeans Clustering, a widely-used unsupervised machine learning algorithm.

Our analysis revealed a strong positive correlation between depression levels and anxiety, as well as stress levels. Additionally, we observed a positive correlation between depression levels and gender, with females exhibiting higher levels of depression, particularly among victims of unwanted contact on social media. Conversely, there was a negative correlation between depression levels and age, as well as education level.



Pattern Identification: For pattern identification, we utilised a [Sentence Transformer model from Hugging Face](#) to evaluate sentence similarity. This facilitated the recognition of recurring patterns within the data, enabling efficient data retrieval when the chatbot receives input.

Overall, by fine-tuning our model for these specific tasks using DistilBERT for Sentiment Analysis and KMeans Clustering for Correlation Analysis and Sentence Transformer model for Pattern Identification, we aimed to optimise its performance and ensure it effectively addresses the challenges associated with these tasks in real-world applications.

RAG pipeline: In our conversational AI system, we've implemented a sophisticated RAG (Retrieval-Augmented Generation) pipeline to enhance the quality of responses provided to users.

- **Retriever:** In the retrieval stage of our RAG pipeline, we've implemented a parent-child retriever mechanism. This advanced retrieval approach leverages hierarchical relationships between documents, allowing us to efficiently retrieve pertinent information from a large corpus of data.
- **Generator:** For the generator component, we utilised the [Mistral 7b instruct v0.2](#) model. This model, trained on conversational data, is adept at generating human-like responses based on the retrieved information.

By incorporating both the parent-child retriever and Mistral 7b instruct v0.2 model into our RAG pipeline, we aimed to improve the relevance and coherence of generated responses to user queries.

Model Fine Tuning: We fine-tuned our system using Structured Fine-Tuning (SFT) and aligned it using Direct Preference Optimization (DPO). First, we formatted the SFT data and generated Synthetic DPO Data. We then fine-tuned the Raw Model using this data and further aligned the SFT Model to the DPO Model. This iterative process involved refining and optimising the models based on the specific data and tasks involved, ultimately enhancing their performance and effectiveness.

For our Supervised Fine Tuning (SFT) process, we employed the Mistral-7B-Instruct-v0.2 BNB 4-bit model. Using unsloth, we were able to perform supervised fine-tuning of the Mistral 7B Instruct v0.2 BNB 4-bit model on Google Colab for 2 epochs. After this, the model seemed to provide better responses. Due to a lack of GPU resources, more epochs could not be trained. We used an LORA adapter with $r=16$. The dataset used had columns instruction and output, with questions and answers and was picked up from Hugging Face and pre-processed before use.

For both SFT and DPO, we utilised configurations such as loraconfig, training arguments, and the respective trainers to train the models. During the DPO process, we injected two matrices into LORA and fine-tuned the Mistral architecture, optimising our system for the specific task at hand.

VectorDB: We utilised [ChromaDB](#) as a vector database for ingesting CSV and PDF files. This allowed us to efficiently store and manage the vector representations of the documents extracted from these files, enabling fast and effective retrieval of information for various tasks such as search and analysis.

LlamaGuardrail: We integrated LlamaGuard to ensure safety, ethical behaviour, and non-toxic interactions within our system. Utilising the [LlamaGuard 7b](#) model, our system classifies user input as safe or unsafe. In instances of unsafe input, the model further categorises it into specific unsafe categories such as Violence & Hate, Sexual Content, Guns & Illegal Weapons, Regulated Substances, Suicide & Self Harm, and Criminal Planning. This proactive approach enhances user protection and fosters a secure and respectful environment within our platform. If a prompt is deemed unsafe, the chatbot will not respond; otherwise, it will engage in normal conversation. Additionally, we implemented a custom category to mitigate risks such as SQL injections and jailbreak attempts, bolstering our system's security measures.

UI and Integration

In our endeavour to integrate our chatbot model effectively, we meticulously followed a series of steps tailored to ensure seamless integration and optimal performance.

Embedding & Database: We utilise Sentence Transformers for generating embeddings from text data then we crafted a loader script to handle the reading of text data, transformation into embeddings using Sentence Transformers, and insertion of these embeddings into your vector database. Finally, run the loader script to seamlessly populate your vector database with the generated embeddings.

Data Retrieval Chain: We employed a conversation chain to retrieve specific instances of data from the vector database, seamlessly navigating through the stored embeddings to access relevant information.

Conversation Prompt & History: For the prompt, we provided a snippet resembling that of a mental health therapist, fostering a supportive and understanding tone conducive to

conversations about mental well-being. Utilising the chat history, we maintained continuity in the conversation, allowing for seamless follow-ups and ensuring that each interaction builds upon previous exchanges. Additionally, we developed prompt templates for the system, human, and AI responses, considering chat history and previous context. These templates were further modified to address issues such as jailbreaking and adversarial prompting.

We utilised a Chatbot Conversation Chain with BufferMemory to facilitate communication between user LLM and the chat history. This involved implementing a Conversational RetrievalChain, complete with a retriever for fetching pertinent information, integrating chat history to preserve context, and incorporating prompts to facilitate the generation of follow-up questions or responses. Additionally, we used ConversationRetrievalChain to integrate RAG (Retrieval-Augmented Generation) capabilities, enhancing the chatbot's ability to provide relevant and informative responses. This comprehensive approach ensured a smooth and coherent conversational experience. Furthermore, this system will be accessed for sentiment analysis, pattern identification, and correlation analysis, further enhancing its utility and analytical capabilities.

Llama Guard Integration: In our implementation, we've established a separate .py file where we interact with the Hugging Face API to access llama guard. After updating specific categories within llama guard, we prompt it for evaluation. Then, within our llm generator file, we integrate llamaguard to assess the safety of user responses, ensuring that only safe content is generated.

Chatbot & Recommendation Integration: We integrated our Chatbot separately from the recommendation system, employing a ConversationChain that includes ConversationBufferMemory. This architecture allows for effective management of dialogue history and context, enhancing the conversational experience for users interacting with the Chatbot.



UI: We integrated our models hosted on Hugging Face Spaces with a user-friendly interface using Streamlit, allowing seamless interaction for users through their web browsers.

Model Deployment

We deployed our application into Hugging Face Spaces, utilising the platform's resources to host and share our trained models seamlessly. First, we ensured our model was prepared

and saved in a compatible format. After creating a Hugging Face account and accessing Spaces, we uploaded our model, providing relevant metadata for easy discovery. Testing was conducted to verify the model's performance within the Hugging Face environment. Additionally, we packaged any necessary components, such as code or data processing scripts, and uploaded them to our Space. Sharing was made simple by providing others with the link to our Hugging Face Space or directly to our deployed model. Regular monitoring and updates ensure our application remains optimised and effective.

Conclusion

The Omdena Hyderabad Local Chapter conducted a comprehensive project analysing social media data related to mental health and developing chatbot tools for mental health support. Through meticulous data collection and preprocessing from diverse sources gathered valuable insights. Advanced techniques such as sentiment analysis and correlation analysis were employed to uncover significant patterns and trends in mental health data. Leveraging various models further enhanced response quality and system performance. Integration of VectorDB and LlamaGuard ensured efficient data storage and system safety, while Streamlit facilitated a user-friendly interface. Overall, the project's multidimensional approach demonstrates a robust framework for leveraging AI in mental health analysis and support.

Future Directions

- Introduce a user feedback feature
- Upgrade from LLM to Fine-Tuned LLM for improved performance
- Implement models for enhanced sentiment prediction
- Identify conversation endpoints
- Comprehend user patterns
- SFT and DPO integration into the chatbot
- Prioritise integration of speech-to-text and text-to-speech capabilities
- Implement language translation functionalities
- Seamlessly incorporate advanced AI features including LLMs, multimodal techniques, and predictive modelling

Links

Dagshub repo:

https://dagshub.com/Omdena/HyderabadIndiaChapter_MentalHealthWellbeingFomoSocialMedia/src/team-5

Application:

https://huggingface.co/spaces/omdena-mhc-team5/mhc_brighterdays_mentor

Collaborators

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